

Rechnerarchitektur - UE8

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Aufgabe 3

a.) FiFo

t	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
A	0xC	0xD	0xA	0x7	0xB	0xB	0xD	0x5	0x4	0x5	0x9	0xA	0xD	0x4	0xC
0	0xC				0xB	0xB					0x9				0xC
1		0xD					0xD					0xA			
2			0xA					0x5		0x5			0xD		
3				0x7					0x4					0x4	
Hit Miss	Miss	Miss	Miss	Miss	Miss	Hit	Hit	Miss	Miss	Hit	Miss	Miss	Miss	Hit	Miss

- Anzahl Misses: 11

b.) LRU

t	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
A	0xC	0xD	0xA	0x7	0xB	0xB	0xD	0x5	0x4	0x5	0x9	0xA	0xD	0x4	0xC
0	0xC				0xB	0xB					0x9				0xC
1		0xD					0xD					0xA			
2			0xA					0x5		0x5				0x4	
3				0x7					0x4				0xD		
Hit Miss	Miss	Miss	Miss	Miss	Miss	Hit	Hit	Miss	Miss	Hit	Miss	Miss	Miss	Miss	Miss

- Anzahl Misses: 12

c.)

Durchschnittliche Zugriffszeit

$$T_{\text{avg}} = \text{Trefferrate} \cdot T_{\text{hit}} + \text{Missrate} \cdot (T_{\text{hit}} + T_{\text{miss}})$$

- FiFo

$$T = 0.92 \cdot 15 + 0.08 \cdot (15 + 250)$$

$$T = 13.8 + 21.2$$

$$T = 35.0 \text{ ns}$$

- **LRU**

$$T = 0.97 * x + 0.03 * (x + 250)$$

$$T = 0.97 * x + 0.03 * x + 7.5$$

$$T = x + 7.5$$

- **Maximal mögliche Zugriffszeit bei LRU - Cache**

$$T = x + 7.5 < 35$$

$$T < 27.5 \text{ ns}$$
